Tengyue Zhang

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Education

Shanghai Jiao Tong University (SJTU), Shanghai, China 2023 – Present

B.S. in Physics (Major) & Computer Science (Minor) (Expected Jun. 2027)

- Member of the **Zhiyuan Honor Program** (Physics Track, Top 10% of Cohort)
- **Overall GPA:** 4.00 / 4.30 (Rank: **2** / 29 as of Feb. 2026)

Research Interests

- **AI for Science:** Integrating AI with physical principles to enhance the performance of neural networks or solve applied problems.
- **Multimodal Generation:** Developing advanced generative models (e.g., Discrete Diffusion) for multi-modal synthesis, with a specific focus on Text-to-Speech (TTS) currently.

Research Experience

ReThinkLab, SJTU Sept. 2025 – Present

Research Assistant Advisor: Prof. Junchi Yan

Conducting research on AI for Science. Key focuses include Quantum Machine Learning (QML) for variational quantum circuits, Physics-Informed Neural Networks (PINNs) for solving PDEs, and AI for Fusion.

X-LANCE Lab, SJTU Jun. 2025 – Present

Research Assistant Advisor: Assoc. Prof. Xie Chen

Developing DiTAR-like TTS model using LocDiT architecture with patchification techniques and discrete diffusion methods based on F5-TTS framework.

Zhiyuan Scholar Program, SJTU Jan. 2025 – Jun. 2027 (Expected)

Research Assistant Advisor: Prof. Hao Zheng

Investigating the spectroscopy of Higher-Order Topological Matter (HOTIs and HOTSCs) on STM platform.

Research Publications

Co4ICF: Co-evolving Physics-Informed Surrogate and RL-based Pulse Optimizer for Inertial Confinement Fusion Feb. 2026

Jiatong Zhao, **Tengyue Zhang***, Yuhan Wang, Fuyuan Wu, Junchi Yan* In Proceedings of The 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26)

Proposed a framework that evolve a PINN surrogate and PPO for ICF laser pulse optimization with bi-frequency calibration to mitigate OOD distribution shift. Achieved +46.1% yield gain under MULTI verification and about 990x optimization speedup.

Skills

- **Dev Tools:** `Python` > `C++` > `Julia`. Fluent at `git`, LaTeXand `Docker`; `Ubuntu` user.
- **Machine Learning:** Self-studied Machine Learning, Deep Learning, Natural Language Processing. Now studying DRL(UCB CS294) and post-training techniques.
- **Generative Models:** Familiar with denoising diffusion models such as DDPM, Flow Matching and Masked Diffusion Model (MDM); Discrete tokenization methods such as VQ-VAE, FSQ, RVQ, etc.
- **Mathematics & Physics Basis:** Achieved A(+) across all CS and physics-related courses (until Feb. 2026).

🏆 Honors and Awards

Meritorious Winner, Interdisciplinary Contest in Modeling (ICM)	May 2025
Huatai Securities Technology Scholarship, Shanghai Jiao Tong University	2024